

DATA SCIENCE CAPSTONE PROJECT REPORT



Executive Summary

This capstone project analyzed SpaceX Falcon 9 launch data to identify the factors influencing first-stage landing success and to develop predictive models for future launch outcomes. The project followed the complete data science methodology as follows :

- data collection,
- data wrangling,
- exploratory data analysis (EDA),
- visualization,
- geospatial analysis,
- dashboard development,
- and machine learning.

Data was collected from multiple sources, including the SpaceX REST API and publicly available web resources through web scraping. The API provided launch, payload, rocket, and landing information, while web scraping supplemented historical launch records and mission details. The collected datasets were cleaned, merged, and transformed into a structured analytical dataset focused on Falcon 9 launches.

During the data wrangling phase, missing values were identified and addressed, irrelevant records were removed, and Falcon 1 launches were excluded to focus exclusively on Falcon 9 missions. Additional features such as launch site, payload mass, orbit type, booster version, landing outcome, and geographical coordinates were extracted and engineered for analysis.

Exploratory Data Analysis (EDA) was conducted using both SQL and Python. SQL Lite queries were used to examine launch frequencies, mission outcomes, launch sites, and orbit distributions. The analysis revealed that launch success rates improved significantly over time, with certain launch sites and mission profiles exhibiting higher landing success probabilities. Payload mass and orbit type were found to have measurable effects on landing outcomes.

Geospatial analysis was performed using Folium to visualize launch site locations and surrounding infrastructure. Interactive maps highlighted the relationship between launch sites, coastlines, transportation networks, and population centers. Launch locations such as Cape Canaveral Space Force Station (CCSFS SLC-40), Kennedy Space Center (KSC LC-39A), and Vandenberg Space Force Base (VAFB SLC-4E) were examined to better understand operational considerations and mission success patterns.

An interactive dashboard was developed to enable dynamic exploration of launch data. Users could analyze launch success rates by site, payload mass, orbit type, and mission outcome. The dashboard provided intuitive visualizations that supported decision-making and facilitated deeper insights into launch performance trends.

Several machine learning classification models were developed to predict Falcon 9 first-stage landing success. Models were trained and evaluated using historical launch data and engineered features. Performance metrics such as accuracy were used to compare model effectiveness. The predictive analysis demonstrated that machine learning can successfully identify patterns associated with landing outcomes and provide valuable forecasts for future missions.

The project demonstrated a complete end-to-end data science workflow and showed that launch site characteristics, payload mass, orbit selection, and mission parameters are important predictors of Falcon 9 first-stage landing success. The results highlight the value of data-driven approaches in supporting aerospace operations, mission planning, and reusable rocket technology.

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INTRODUCTION

SpaceX has transformed the commercial aerospace industry through the development of reusable rocket technology. The Falcon 9 launch vehicle is designed to recover and reuse its first-stage booster, significantly reducing launch costs compared to traditional expendable rockets. Predicting whether a Falcon 9 first-stage booster will successfully land is valuable because it directly impacts mission economics, operational planning, and future launch scheduling.

The objective of this capstone project was to analyze historical Falcon 9 launch data and identify the factors that influence first-stage landing success. Using data collected from the SpaceX API and publicly available web sources, the project explored launch characteristics, mission outcomes, launch sites, payload information, and orbital destinations. The final goal was to develop machine learning models capable of predicting landing success based on historical mission data.

METHODOLOGY

The project followed a structured data science workflow consisting of six major phases:

Data Collection

Launch data was collected using the SpaceX REST API, which provided information on launches, rockets, payloads, launch sites, and landing outcomes. Additional mission-related information was obtained through web scraping techniques to supplement and validate the dataset.

Data Wrangling

The collected datasets were cleaned and transformed into a structured analytical dataset. Missing values were identified and handled appropriately. Falcon 1 launches were removed to focus exclusively on Falcon 9 missions. Relevant variables including launch site, orbit type, payload mass, landing outcome, booster version, and geographic coordinates were extracted and engineered.

Exploratory Data Analysis (EDA)

Exploratory analysis was performed using Python and SQL. Statistical summaries, visualizations, and SQL queries were used to examine relationships between launch outcomes and mission characteristics. Launch frequencies, orbit distributions, launch site performance, and payload trends were analyzed.

Geospatial Analysis

Folium was used to create interactive maps displaying launch sites and nearby infrastructure. Geographic relationships between launch facilities, coastlines, transportation networks, and population centers were examined to better understand operational environments.

Dashboard Development

An interactive dashboard was developed to visualize launch success rates, payload distributions, orbit classifications, and launch site performance. The dashboard enabled users to explore trends dynamically through filtering and interactive charts.

Predictive Modeling

Several machine learning classification algorithms were trained and evaluated using historical Falcon 9 launch data. Feature engineering and preprocessing techniques were applied before model training. Model performance was assessed using standard classification metrics and cross-validation techniques.

RESULTS

The analysis produced several important findings:

- A total of approximately 90 Falcon 9 launches were analyzed after removing Falcon 1 missions.
- Launch success rates improved significantly over time, reflecting SpaceX's growing experience with reusable launch systems.
- The most active launch site was CCSFS SLC-40, followed by KSC LC-39A and VAFB SLC-4E.
- Geostationary Transfer Orbit (GTO) missions represented a significant portion of launches and were associated with varying landing success rates due to mission complexity.
- Payload mass demonstrated a measurable relationship with landing outcomes. Heavier payloads generally required more fuel expenditure, influencing recovery performance.
- Launch site selection and orbit type were among the strongest operational variables associated with successful booster recovery.
- Geospatial analysis confirmed the strategic positioning of launch sites near coastlines and transportation infrastructure, supporting efficient launch and recovery operations.
- Machine learning models achieved strong predictive performance in classifying landing outcomes. Classification algorithms successfully identified patterns linking mission characteristics to booster recovery success.

DISCUSSION

The findings demonstrate that Falcon 9 landing success is influenced by a combination of technical and operational factors. Launch site characteristics, payload mass, mission orbit, and booster configuration all contribute to recovery outcomes.

The analysis confirms the effectiveness of SpaceX's reusable launch strategy. Historical trends indicate continuous improvement in landing reliability, supporting the company's objective of reducing launch costs through booster reuse.

The dashboard and geospatial analyses provided valuable tools for visualizing mission patterns and identifying operational trends. Additionally, the machine learning models demonstrated that historical launch data can be used effectively to predict future landing outcomes.

Several limitations should be noted. The dataset is limited to publicly available information and historical missions. Future models could potentially be improved by incorporating additional engineering variables such as weather conditions, fuel margins, and real-time flight telemetry.

CONCLUSION

This project successfully applied the complete data science methodology to analyze SpaceX Falcon 9 launch operations and predict first-stage landing outcomes.

Through data collection, wrangling, exploratory analysis, geospatial visualization, dashboard development, and machine learning, key factors affecting landing success were identified. Launch site, payload mass, orbit type, and mission profile were found to be significant predictors of successful booster recovery.

The predictive models demonstrated that machine learning techniques can accurately forecast Falcon 9 landing outcomes using historical launch data. These findings illustrate the practical value of data science in aerospace operations and support the continued advancement of reusable rocket technologies.

APPENDIX

Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Folium
- SQL Lite
- Jupyter Notebook
- Plotly Dash
- Scikit-learn
- SpaceX REST API
- BeautifulSoup (Web Scraping)

Dataset Sources

- SpaceX REST API
- Public SpaceX Mission Records
- Launch Site Geospatial Data

Machine Learning Models Evaluated

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree Classifier
- K-Nearest Neighbors (KNN)

Project Deliverables

- Data Collection Notebook
- Data Wrangling Notebook
- SQL Analysis Notebook
- Interactive Dashboard
- Folium Launch Site Map
- Machine Learning Prediction Notebook
- Final Presentation
- Final Report